

Graded Where-Clauses over the Reals

One-dimensional grading as the mortal scientist's gradient,
and probabilistic logic as the metric that gradient requires

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Paper II of three; supersedes the $\mathbb{R}_{\geq 0}$ half of *Graded Where-Clauses*

Abstract

The rho calculus does not say how the non-determinism of a race is resolved. The companion papers treat that silence as a parameter and instantiate it: with $\mathbb{V} = \mathbb{C}$ one obtains amplitudes and the Born rule. Here $\mathbb{V} = \mathbb{R}_{\geq 0}$, and the question is not what a program means but what a *learner* can do with a value it is allowed to tune.

The answer turns on one structural fact. A learner holding a graded hypothesis has two degrees of freedom, and they live in different geometries. The formulae form an ultrametric space, in which there is no betweenness and therefore no gradient; the values form an ordered continuum, in which there is nothing but gradient. So the hypothesis space is a fibration, and one-dimensional grading is exactly the enrichment that makes the fibre navigable.

We make the fibre gradient precise. Over $\mathbb{R}_{\geq 0}$ a modality-free clause is a posynomial in its scalar parameters, hence smooth and log-convex in log-coordinates; the likelihood of a causal history factors over firing sites; and the parameter gradient of that likelihood decomposes into a sum of local scores, each computable at the site from the candidate values the site already had. Differentiating a *modal* clause is the adjoint of the same transfer operator whose Neumann series defines the graded fixed point, so differentiability and existence hold under one condition rather than two. None of this survives the passage to $\mathbb{C}$, which is the structural reason the learning instance and the quantum instance are separate papers.

The headline result is that the metric on the fibre is not free, and that choosing it correctly yields probabilistic logic rather than requiring it. The Euclidean gradient on a strength in $[0, 1]$ does not know how much evidence backs the current value; the natural gradient does, through the Fisher information, and the Fisher information is a count. Revising a PLN truth value (s, n) by one observation is *exactly* the natural-gradient step with unit step size and the metric evaluated at the posterior count — measured worst discrepancy 1.110×10^{-16} over 20,000 random instances, against 4.612×10^{-1} if the metric is taken at the prior count. Because the metric is evaluated after the increment the step is implicit, hence unconditionally stable: over 200,000 random steps it never leaves the unit interval, while the explicit forms are absorbed at the boundary or escape it outright. Confidence is therefore not a second truth value. It is the metric on the first.

The consequences are ecological. The confidence schedule satisfies the Robbins–Monro conditions for every value of PLN's personality parameter k , so k governs the transient and not the limit — and only a mortal learner can be harmed by a transient. We measure the optimum for k across the three distributions that specify an ecology: over trophic types, over inedibles, and over token reservoirs housed in no computation. Under expected yield and under log wealth the optima differ systematically and by a factor of seven at the baseline, and the two criteria disagree not only about the personality but about whether learning is worth anything at all. Finally, two learners sharing a belief namespace exhibit the failure the conservation laws predict: evidence is not conserved, revision presumes independence, and circulating testimony inflates the count without adding information.

In gradient terms the inflated count is a collapsed step size, so the pathology of minted confidence is not recklessness but rigidity.

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1 Introduction

1.1 Where the choice is

A rho term may offer several ways to take a step. Two messages waiting on one channel and a single receipt listening there is the smallest example: either message may be consumed, the calculus says both are possible, and it says nothing about which. An implementation must decide, and the decision is not in the language. In the classical implementation it is made by the physical races of cores against a tuple space; in the companion paper on the complex instance it is made by the Born rule.

The design shared by all three papers is one line of grammar. The **where** clause of a for-comprehension, which in rho 1.4 is a Boolean guard, is allowed to take values in a commutative semiring $\mathbb{V}$, and is evaluated once per candidate match rather than once per receipt. When $\mathbb{V} = \mathbb{B}$ nothing changes. When $\mathbb{V}$ is something else, the clause is a weight on a branch, written in the program, by the programmer, at the point where the branching happens.

This paper takes $\mathbb{V} = \mathbb{R}_{\geq 0}$.

1.2 The thesis

The reason to care about $\mathbb{R}_{\geq 0}$ is not that it gives a stochastic semantics. Weighted transition systems are old, and a semiring-valued weight on a transition is not by itself news. The reason to care is that $\mathbb{R}_{\geq 0}$ is *one-dimensional and ordered*, and that a learner whose hypotheses are formulae has, without it, nowhere to put a gradient.

The mortal scientist of the companion programme holds hypotheses drawn from a logic generated by the term structure of rho. Those formulae are metrised by modal depth: two formulae agreeing to depth n are within 2^{-n} . That metric is an ultrametric, and an ultrametric has no betweenness — for distinct φ and ψ there is no formula standing partway between them in the sense a convex combination would. A gradient presupposes a direction one may move along by an arbitrarily small amount. The hypothesis lattice supplies branching instead, so

the only search available on it is tree search, and a theorem of confinement says an incremental walk never leaves the ball it began in.

Grading changes this, but not by repairing the lattice. It adds a second coordinate that lives somewhere else:

A graded hypothesis is a pair, a formula and a value. Changing the value is a move of distance zero in the ultrametric. So the hypothesis space is not a metric space with a decoration; it is a fibration, with the formula lattice as base and a $\mathbb{V}$ -valued fibre over each point — and the fibre is where gradient descent lives.

That much is stated in the omnibus. What it leaves open is everything a learner would need in order to act on it. Which gradient? Of what objective? Computed where, and by whom? Is it available at all when the clause carries a modality? And, given that a value in $[0, 1]$ can be pushed in a direction, what is the right size of push?

The last question is the one with the interesting answer, and it is the organising claim of this paper. The naive gradient on a strength is wrong in a specific and diagnosable way: it does not know how much evidence stands behind the value it is moving. Correcting it requires a metric on the fibre, the metric is the Fisher information, the Fisher information for a Bernoulli parameter is an evidence count, and a strength together with an evidence count is a PLN simple truth value. Probabilistic logic is not adopted here as an application. It is what one gets by asking for the gradient to be taken correctly.

1.3 Contributions

1. A definition of the parameters of a graded clause, and the observation that over $\mathbb{R}_{\geq 0}$ a modality-free clause is a posynomial in them: smooth everywhere on the positive orthant, and log-convex in log-coordinates (§5, Proposition 5.2). Its normalisation is not log-convex, so the fibre is smooth but not convex; both facts are measured.
2. A decomposition theorem: the parameter gradient of the log-likelihood of a causal history is the sum of local scores, one per firing site, each computable from the candidate weights that site already evaluated (Theorem 5.6). The causal-history quotient of the companion paper is what makes the statement well posed, and the estimator is exactly the likelihood-ratio estimator of reinforcement learning, obtained here from the operational semantics rather than assumed.
3. The modal case, worked. The derivative of a graded fixed point is the solution of the adjoint system $(\text{Id} - T) \partial v = \partial b + (\partial T)v$, whose Neumann series is the series that defines the fixed point itself, so differentiability and existence are one condition (Theorem 5.12).
4. **The headline.** PLN revision is a natural-gradient step on the Bernoulli log-likelihood, with unit step size and the Fisher metric evaluated at the posterior count (Theorem 6.3), and the placement of the metric after the increment makes the step implicit, hence unconditionally stable (Corollary 6.5). Confidence is the metric, not a second value.
5. The confidence schedule is a Robbins–Monro schedule for every personality parameter (Proposition 7.2), so k is a transient parameter, and the ecology can price what statistics cannot.
6. A robustness study of that price over the three distributions that specify an ecology — trophic types, inedibles, and free reservoirs — reported under two objectives side by side (§9).
7. A two-learner testimony experiment showing that circulating belief inflates the evidence count without adding information, and that the resulting pathology is loss of plasticity rather than instability (§11).

8. A negative result that justifies the split of this line of work into separate papers: none of the above is available over $\mathbb{C}$, because there is no per-site likelihood to differentiate (Remark 3.6).

1.4 What this paper does not do

It does not propose a new logic; the hypothesis logic is the one generated by the term structure, taken as given. It does not claim an empirical result about agents that use probabilistic logic: the ecology of §9 is a constructed decision problem, and what is reported is the shape of a robustness surface over it, not a general fact about learners. It does not implement any of this in a running node. And it does not treat the complex instance, which is Paper I, or the resolution algebra in general, which is Paper III.

2 The calculus

The presentation is self-contained. Readers of Paper I will find §2.1 and §2.3 familiar, minus the quantum namespace, which plays no part here.

2.1 Terms

Definition 2.1 (Syntax). Processes P and names x are given by

$$\begin{aligned} P &::= 0 \mid x!(P) \mid \text{for}(\vec{y} \leftarrow \vec{x} \text{ where } \psi) P \mid P \mid P \mid *x \\ x &::= @P \end{aligned}$$

A receipt for $(y_1 \leftarrow x_1 \& \cdots \& y_m \leftarrow x_m \text{ where } \psi)P$ is a *join*: it fires only when every one of its m patterns is matched. The clause ψ is drawn from the graded formula sort of §2.2 and takes values in $\mathbb{V}$. Writing the clause is optional; its absence abbreviates $\top$.

Definition 2.2 (Structural congruence). $\equiv$ is the least congruence making $(\mid, 0)$ a commutative monoid and satisfying $*@P \equiv P$ and $@*x = x$.

Remark 2.3 (Receipts are linear). A receipt is consumed when it fires. The persistent form, written $\Leftarrow$, survives; it is available but is not the default, and the distinction matters here for the same reason it matters in Paper I. Under a persistent receipt the two-message configuration of §1.2 is not a race at all: both messages are eventually consumed, the receipt survives either way, and the two derivations reconverge. It is linearity that makes the surviving message a record of which branch was taken.

Definition 2.4 (Candidates). Fix a term P and a receipt occurrence r in it with patterns $y_1, \dots, y_m$ on channels $x_1, \dots, x_m$. A *candidate* at r is an injective assignment σ of the patterns of r to messages resting on the corresponding channels. $\text{Cand}(P, r)$ denotes the set of candidates. Two candidates *contend* if they share a message or share a linear receipt; a *contention set* is a connected component of the contention relation, and it is the unit at which weights are normalised.

2.2 Two sorts of formula

The logic generated by Definition 2.1 has one structural connective per term former, a name predicate, and one behavioural modality per rewrite rule together with a choice of redex position. We keep two sorts.

Definition 2.5 (Formula sorts). The *crisp* sort β retains the full propositional structure the target specification supplies, including negation and fixed points where they exist:

$$\beta ::= \top \mid \text{out}(n)\beta \mid \text{in}(n)\beta \mid \beta \mid \beta \mid @\beta \mid \beta \wedge \beta \mid \neg\beta \mid \langle K \rangle \beta \mid \nu X.\beta.$$

The *graded* sort ψ has no negation, and takes values in $\mathbb{V}$:

$$\psi ::= \lceil \beta \rceil \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle_{\mathbb{V}} \psi \mid X \mid \mu X.\psi \mid \nu X.\psi,$$

where v ranges over constants of $\mathbb{V}$ and fixed-point variables occur positively and guarded. The embedding $\lceil - \rceil$ sends a crisp formula to the multiplicative unit where it holds and to zero where it does not.

The two sorts divide the labour cleanly: the crisp sort decides *whether* a candidate is admissible at all, and the graded sort decides *how much*. The absence of negation on the graded sort is not an oversight. Over a general semiring there is no canonical complement, and over $\mathbb{R}_{\geq 0}$ in particular there are no additive inverses; §3.2 shows that this absence is precisely what makes the present instance the learning instance.

2.3 Per-match evaluation

Definition 2.6 (The weighted communication rule). Let r be a receipt occurrence in P and $\sigma \in \text{Cand}(P, r)$ a candidate. Write $\psi[\sigma] \in \mathbb{V}$ for the value of the clause with the pattern variables bound as σ prescribes. If $\psi[\sigma] \neq 0$ then

$$P \longrightarrow_{\psi[\sigma]} P'$$

where P' is the residue of consuming the assigned messages and r , and substituting. A candidate whose clause value is 0 offers no transition.

Two things about this definition carry weight later. First, evaluation is *per candidate*: the clause sees the bindings, so it may depend on the data, which is what allows it to be a policy rather than a constant. Second, the guard evaluator is pure — in the reference implementation it is enforced by construction, the evaluator being a separate crate that cannot perform effects — so evaluating a clause on a candidate that does not go on to fire costs tokens but changes nothing else. Without that, a learner could not compare its options without altering them.

Remark 2.7 (Plasticity is free). Because the clause is evaluated against the current term, and because beliefs may sit in the term as data on a channel, a clause whose value depends on a belief changes its value when the belief is revised. No update function, no separate sub-language for updates, and no schedule: the machinery an earlier note needed in order to let weights change during execution is subsumed by the fact that a formula is evaluated where it stands.

3 The resolution algebra over the non-negative reals

3.1 The algebra

Definition 3.1 (Resolution algebra). A resolution algebra is a commutative semiring $(\mathbb{V}, \oplus, \otimes, 0, 1)$. Throughout this paper $\mathbb{V} = (\mathbb{R}_{\geq 0}, +, \times, 0, 1)$.

Conjunction of graded formulae multiplies and disjunction adds. A join therefore multiplies the weights of its conjuncts, which is the right reading: the candidate must satisfy all of them, and independent evidence composes multiplicatively. Disjunction adds, which is where two ways of being right reinforce one another.

Definition 3.2 (Completion). A contention set is *complete* when the weights of its maximal matchings sum to 1, and *substochastic* when they sum to at most 1. Resolution normalises within a contention set: the probability that a given matching fires is its weight divided by the total over the set.

Remark 3.3 (Gillespie, precisely). Normalised selection among enabled candidates gives the embedded *jump chain* of a continuous-time Markov chain, not the chain itself. The waiting-time law, in which the total enabled propensity governs an exponential holding time, is a further piece of structure and it is supplied by the weighted-GSLT companion note, where propensities carry time units. Where this paper says a graded program over $\mathbb{R}_{\geq 0}$ is a stochastic simulation, it means the jump chain. The distinction is not pedantry: a learner tuning clause values is tuning the jump chain, and the holding times it does not touch are exactly the part of the dynamics its gradient cannot reach.

3.2 Support adequacy, and why this is the learning instance

Proposition 3.4 (Support adequacy is exact over $\mathbb{R}_{\geq 0}$). *Let $\text{Reach}(P)$ be the set of terms reachable from P under the unweighted semantics, and $\text{supp}[[P]]$ the set carrying non-zero weight under Definition 2.6. Then $\text{supp}[[P]] \subseteq \text{Reach}(P)$ always, and over $\mathbb{R}_{\geq 0}$ with all clause constants strictly positive the inclusion is an equality.*

Proof. The inclusion holds because a weighted step is a step. For the converse, a reachable term is reached by some sequence of candidates; each contributes a strictly positive factor, and a sum of strictly positive numbers is strictly positive, so no cancellation can remove it. Over a semiring with additive inverses the last step fails. $\square$

This is a small proposition with a large consequence, and it is worth stating what it buys.

Principle 3.5 (No cancellation, hence a likelihood). Over $\mathbb{R}_{\geq 0}$ every reachable outcome carries positive weight, so normalisation yields a genuine probability for every branch of every contention set; a derivation therefore has a likelihood, and a likelihood may be differentiated. Over $\mathbb{C}$ a reachable outcome may carry weight zero — that is what interference is — so there is no per-branch probability to take the logarithm of, and the whole apparatus below is unavailable.

Remark 3.6 (The split, justified structurally). It is tempting to read the division of this work into a complex paper and a real paper as editorial convenience. It is not. The complex instance is the one where the semantics is interesting and the learner is impossible; the real instance is the one where the semantics is routine and the learner is the whole point. The technical hinge is a single algebraic fact, the presence or absence of additive inverses, and it decides the presence or absence of a gradient.

4 The hypothesis space is a fibration

4.1 What is imported

The learner in question is the mortal scientist of the companion programme. The minimum needed here is this. A scientist is a cost-accounted ρ computation with a finite token stack. It forms hypotheses in the logic generated by the term language, tests them by placing them in the **where** clause of a receipt — an *assay* — and pays for each test at a price set by the modal depth of the formula. It eats: tokens sit in stacks, some of them inside other computations, and breaking one open is the analogue of predation. It dies when the stack runs out. Everything about its epistemology is therefore priced, and the prices are in the same currency as its survival.

Two results about its hypothesis space are load-bearing here.

Proposition 4.1 (The geometry is a tree). *Formulae agreeing to modal depth n are within 2^{-n} . The resulting metric is an ultrametric: balls are nested or disjoint, every point of a ball is a centre of it, and there is no betweenness.*

Proposition 4.2 (Confinement). *If $\varphi_0, \dots, \varphi_j$ is a revision walk with $d(\varphi_{i-1}, \varphi_i) < 2^{-n}$ for every i , then $d(\varphi_0, \varphi_j) < 2^{-n}$.*

Confinement is why the companion note says gradient methods are not merely a poor fit for hypothesis revision but are not well typed: a gradient needs a direction in which one may move by an arbitrarily small amount, and an ultrametric supplies branching instead. That verdict is correct, and this paper does not overturn it. It locates the gradient somewhere else.

4.2 Two degrees of freedom

Definition 4.3 (Graded hypothesis). A *graded hypothesis* is a pair (ψ, θ) where ψ is a graded formula and $\theta \in \mathbb{R}_{\geq 0}^d$ assigns values to the d constant symbols occurring in it. Two graded hypotheses have the same *shape* when they differ only in θ .

Proposition 4.4 (The hypothesis space is a fibration). *Revising θ leaves the shape, hence the formula, unchanged, and is therefore a move of ultrametric distance zero. The space of graded hypotheses is the total space of a fibration whose base is the ultrametric lattice of shapes and whose fibre over a shape is $\mathbb{R}_{\geq 0}^d$.*

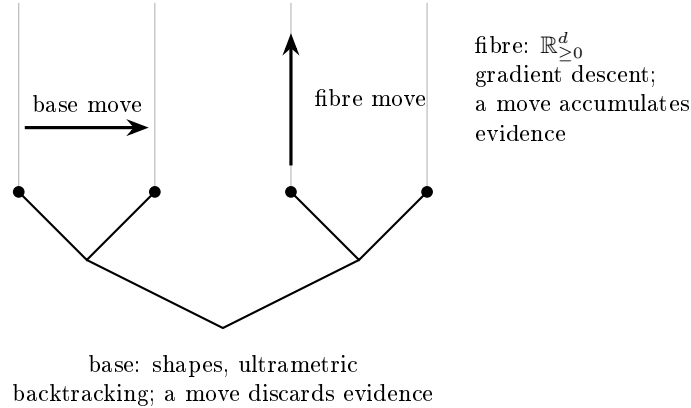


Figure 1: Confinement is a theorem about the base. A fibre move has base-distance zero, so it cannot leave a ball — and can go anywhere inside one.

Principle 4.5 (Two searches, one learner). The base admits only backtracking search, priced by how far one had to back up; the fibre admits gradient search, priced by the assays that produced the evidence. A base move discards confirmations obtained below the depth of the locus; a fibre move accumulates them. A learner needs both, and the two are not interchangeable.

Remark 4.6 (What grading does not do). It does not escape confinement. A learner that only revises values never leaves the point it started at, let alone the ball. The forager of §9 is the illustration: it learns not to open, which saves its life, and it never acquires a better test. Symmetrically, a learner that only moves in the base can acquire deeper theories but pays for each, and — as the companion note measures — may be made worse by depth while holding its other commitments fixed. Both pathologies are visible in numbers, and neither is visible without the other coordinate.

5 The gradient exists

We now make Proposition 4.4 operational. Three things are required: that the fibre be smooth, that there be an objective whose gradient is wanted, and that the gradient be computable by the learner rather than by an observer of it.

5.1 Clauses are posynomials

Definition 5.1 (Parameters). The *parameters* of a graded clause are the constant symbols of $\mathbb{V}$ occurring in it. A clause with parameters $\theta = (\theta_1, \dots, \theta_d)$ and a candidate σ determine a value $w(\theta, \sigma) \in \mathbb{R}_{\geq 0}$.

Proposition 5.2 (Modality-free clauses are posynomials). *Let ψ be modality-free and fixed-point-free. Then for each candidate σ ,*

$$w(\theta, \sigma) = \sum_{i=1}^N c_i(\sigma) \prod_{j=1}^d \theta_j^{a_{ij}}, \quad c_i(\sigma) \in \{0, 1\}, \quad a_{ij} \in \mathbb{N},$$

a posynomial in θ : the crisp subformulae contribute the coefficients and the graded structure contributes the monomials. Consequently w is smooth on the positive orthant, and $\log w$ is convex as a function of $\log \theta$.

Proof. Induct on ψ . A constant is a monomial; $[\beta]$ is the constant 1 or 0 according to whether β holds at σ , since the guard evaluator is pure and does not see θ ; $\otimes$ is multiplication, which multiplies monomials; $\oplus$ is addition, which concatenates the sums. Smoothness is immediate. For log-convexity, write $\theta_j = e^{u_j}$; each monomial becomes $e^{\langle a_i, u \rangle}$, so w is a positive combination of exponentials of affine functions, and the logarithm of such a combination is convex. $\square$

Proposition 5.3 (Normalisation is not log-convex). *The probability that a candidate fires within its contention set, $p = w_\sigma / \sum_\tau w_\tau$, has $\log p = \log w_\sigma - \log \sum_\tau w_\tau$, a difference of convex functions of $u = \log \theta$. It is in general neither convex nor concave.*

Empirical finding 5.4 (Curvature, measured). Over 400 random modality-free clauses in 3 parameters, evaluated at random points of the positive orthant, the Hessian of $\log w$ in log-coordinates had least eigenvalue -8.882×10^{-8} — zero to the accuracy of the finite-difference scheme, consistent with Proposition 5.2. The Hessian of $\log p$ was indefinite in 366 of the same 400 trials, with most negative eigenvalue -2.603 . Computed by `fibre.py`, experiment E4.

The reading is the honest one. The fibre is smooth, so a gradient exists everywhere on it, and clause values considered alone are as well behaved as a geometric program. What is *not* available is a guarantee that following the gradient reaches a global optimum, because the act of normalising — of turning weights into a choice — destroys convexity. A learner in this framework is therefore in the ordinary position of a gradient learner: local moves, no certificate.

5.2 The likelihood of a causal history

Definition 5.5 (Causal history). A *causal history* h of length ℓ is a sequence of firings $(\Gamma_1, \sigma_1), \dots, (\Gamma_\ell, \sigma_\ell)$, where each Γ_i is a contention set enabled at step i and σ_i a maximal matching of it, quotiented by the interleaving of independent firings.

The quotient is not a hygiene measure. Without it, m independent firings contribute the $m!$ linearisations of a partial order, and the likelihood overcounts by that factor; the companion

papers measure the inflation directly and it is exactly 2, 6, 24 for $m = 2, 3, 4$. With it, the likelihood is well defined:

$$\Pr_\theta(h) = \prod_{i=1}^{\ell} \frac{w(\theta, \sigma_i)}{\sum_{\tau \in \Gamma_i} w(\theta, \tau)}. \quad (1)$$

Theorem 5.6 (The score decomposes over firing sites). *With $\Pr_\theta(h)$ as in (1),*

$$\frac{\partial}{\partial \theta_j} \log \Pr_\theta(h) = \sum_{i=1}^{\ell} \left(\frac{\partial_j w(\theta, \sigma_i)}{w(\theta, \sigma_i)} - \frac{\sum_{\tau \in \Gamma_i} \partial_j w(\theta, \tau)}{\sum_{\tau \in \Gamma_i} w(\theta, \tau)} \right).$$

Each summand depends only on data local to the firing: the candidates of that contention set, their clause values, their parameter derivatives, and which matching fired.

Proof. Take logarithms in (1) and differentiate term by term; each term is a difference of two logarithmic derivatives. Locality is by inspection: no factor mentions a step other than its own. $\square$

Empirical finding 5.7 (The decomposition, measured). Over 300 random histories of 6 contention sets each in 3 parameters, the worst discrepancy between the sum of local scores and a central-difference gradient of the total log-likelihood was 1.221×10^{-9} . `fibre.py`, experiment E5.

Corollary 5.8 (The gradient is a message, not a pass). *A learner can accumulate $\partial_j \log \Pr_\theta(h)$ by adding one number per firing to a running total. No global traversal of the reduction graph is required, and nothing must be stored that the site did not already compute in order to resolve its own race.*

Remark 5.9 (What this is, in other vocabularies). Theorem 5.6 is the likelihood-ratio or score-function estimator; the running total of Corollary 5.8 is an eligibility trace; and combining either with a terminal payoff is policy-gradient reinforcement learning. The point of deriving rather than importing them is that here the policy is not a neural network chosen for differentiability. It is the **where** clause, and its differentiability is a consequence of Proposition 5.2, which is a consequence of the semiring. The learner’s policy and the program’s semantics are the same object, so there is no interface between them to get wrong.

Remark 5.10 (Where the objective comes from). Nothing so far says what should be maximised. The framework answers this without a hyperparameter: the learner has a token stack, the stack is conserved, and running out of it ends the computation. The objective is therefore metabolic. Which functional of the stack, however, is a genuine choice, and §9 reports two — expected terminal stack and expected log terminal stack — side by side, because they disagree.

5.3 The modal case

A clause bearing $\langle K \rangle_{\forall}$ under a fixed point is not a posynomial, and Proposition 5.2 does not apply. This is the case a learner actually wants, because it is the case in which the clause is a hypothesis about the future rather than about the datum in hand. It can be handled exactly.

Definition 5.11 (The transfer operator). Over a finite set of configurations $\{1, \dots, n\}$, the graded modality is the matrix T with T_{pq} the weight of the transition from p to q . The clause $\psi = \mu X. ([\beta] \oplus \langle K \rangle_{\forall} X)$ denotes the least solution v of

$$v = b + Tv, \quad b_p = \text{value of } [\beta] \text{ at } p,$$

which exists and equals $\sum_{i \geq 0} T^i b$ when the spectral radius satisfies $\rho(T) < 1$.

Theorem 5.12 (The adjoint gives the gradient). *Suppose b and T depend smoothly on θ and $\varrho(T) < 1$. Then v depends smoothly on θ and*

$$(\text{Id} - T) \partial_j v = \partial_j b + (\partial_j T) v, \quad \text{hence} \quad \partial_j v = \sum_{i \geq 0} T^i (\partial_j b + (\partial_j T) v).$$

The series on the right is the Neumann series of the same operator that defines v . It converges under exactly the hypothesis that makes v exist.

Proof. Differentiate $v = b + Tv$ with respect to θ_j and collect: $\partial_j v = \partial_j b + (\partial_j T)v + T\partial_j v$. Since $\varrho(T) < 1$, $\text{Id} - T$ is invertible with inverse $\sum_i T^i$, which gives both the solution and its convergence; smoothness is the implicit function theorem applied to $F(\theta, v) = v - b(\theta) - T(\theta)v$, whose partial derivative in v is $\text{Id} - T$, invertible by hypothesis. $\square$

Corollary 5.13 (One condition, not two). *A modal graded clause over $\mathbb{R}_{\geq 0}$ is differentiable in its parameters exactly where it is defined. There is no separate regularity hypothesis for learning.*

Empirical finding 5.14 (The adjoint, measured). Over 200 random systems of dimension 5 in 2 parameters with spectral radius ranging over $[0.133, 0.952]$, the worst discrepancy between the adjoint solution and a central-difference derivative of the fixed point was 2.933×10^{-8} . Truncating the Neumann series at 60 terms gives, at spectral radius 0.50, 0.90, 0.99, 1.00 and 1.05, the values 2.000000, 9.982030, 45.284336, 60.000000 and 353.583718 against exact values 2.000000, 10.000000, 100.000000 and divergence in the last two: the approach to the boundary degrades value and gradient together. `fibre.py`, experiment E6.

Remark 5.15 (Backpropagation, and what it is here). Theorem 5.12 is reverse-mode differentiation through a fixed point, and the practical recipe it licenses — iterate the adjoint to the same tolerance one iterated the value — is the recipe used for implicit layers in machine learning. What is worth noting is the direction of the derivation. There, one designs a differentiable architecture and then studies its fixed points. Here, the fixed point is what the logic already meant by a formula holding henceforth, and its differentiability is a theorem about the semiring. The learner does not have a differentiable model of its environment. It has a hypothesis, and the hypothesis happens to be differentiable in its parameters.

6 Which gradient? The metric on the fibre

6.1 The naive step is wrong, and wrong in a diagnosable way

Fix the simplest interesting fibre. A learner holds one hypothesis of the shape “a candidate satisfying the cheap structural test χ is worth acting on”, and the clause value on such a candidate is a single parameter $s \in (0, 1)$. Each time the learner acts it observes whether the candidate was in fact worth acting on, an outcome $x \in \{0, 1\}$. The obvious objective is the Bernoulli log-likelihood

$$\ell(s; x) = x \log s + (1 - x) \log(1 - s), \quad \mathcal{S}(s; x) := \frac{\partial \ell}{\partial s} = \frac{x - s}{s(1 - s)}.$$

Gradient ascent on ℓ is $s \mapsto s + \eta \mathcal{S}(s; x)$, and it is a bad move for two reasons that are really one reason. It is unbounded near the ends of the interval, so a confident learner meeting a surprise takes an enormous step. And it is insensitive to how much evidence the current value rests on: a learner that has seen one instance and a learner that has seen a thousand take the same step on the next observation. Both faults are the absence of a metric. The parameter s is a coordinate on a statistical manifold, and the Euclidean geometry of the coordinate is not the geometry of the manifold.

Definition 6.1 (The natural gradient). The natural-gradient step with respect to a Riemannian metric $\mathcal{I}$ on the parameter is $s \mapsto s + \eta \mathcal{I}^{-1} \mathcal{S}$. Taking $\mathcal{I}$ to be the Fisher information of the model is the canonical choice, and for n independent Bernoulli observations it is

$$\mathcal{I}(s; n) = \frac{n}{s(1-s)}.$$

The Fisher information for a Bernoulli parameter is a count divided by a variance. The variance cancels against the score. What is left is a count.

6.2 The headline

Definition 6.2 (PLN simple truth values and revision). A simple truth value is a pair (s, n) with $s \in [0, 1]$ a strength and $n \geq 0$ an evidence count; the associated confidence is $c = n/(n+k)$ for a personality parameter $k > 0$. Revision by a fresh observation of strength x and count m is

$$\text{rev}((s, n), (x, m)) = \left(\frac{ns + mx}{n + m}, n + m \right).$$

Theorem 6.3 (Revision is a natural-gradient step). *For every $s \in (0, 1)$, every $n > 0$ and every x , the strength component of $\text{rev}((s, n), (x, 1))$ equals the natural-gradient ascent step on the Bernoulli log-likelihood with unit step size and Fisher information evaluated at the posterior count:*

$$\frac{ns + x}{n + 1} = s + \mathcal{I}(s; n + 1)^{-1} \mathcal{S}(s; x).$$

Proof. $\mathcal{I}(s; n + 1)^{-1} \mathcal{S}(s; x) = \frac{s(1-s)}{n+1} \cdot \frac{x-s}{s(1-s)} = \frac{x-s}{n+1}$, and $s + \frac{x-s}{n+1} = \frac{ns+x}{n+1}$. $\square$

Empirical finding 6.4 (The identity, measured, with its negative control). Over 20,000 instances with s uniform on $[0.02, 0.98]$, n uniform on $\{1, \dots, 500\}$ and x uniform on $\{0, 1\}$, the worst absolute discrepancy between revision and the natural-gradient step was 1.110×10^{-16} with the Fisher information taken at $n + 1$, and 4.612×10^{-1} with it taken at n . The identification is therefore sharp: it is not merely that revision is gradient-like, but that it is this step and not the neighbouring one. `fibres.py`, experiment E1.

Corollary 6.5 (The step is implicit, hence stable). *Because the metric is evaluated after the increment, the update is the implicit (backward-Euler, or proximal) discretisation of the natural gradient flow. It is a convex combination of s and x with weights $n/(n+1)$ and $1/(n+1)$, so for $n > 0$ and $s \in (0, 1)$ the result lies strictly inside the unit interval for every observation and every prior state. The explicit discretisation does not have this property.*

Empirical finding 6.6 (Stability, measured). Over 200,000 random steps with s uniform on $(0, 1)$ and $n \in \{1, \dots, 50\}$:

discretisation	escapes $[0, 1]$	absorbed	worst excursion
implicit natural gradient (= revision)	0	0	0.000
explicit natural gradient, step 1	0	3,882	0.000
explicit natural gradient, step 1.5	3,882	0	0.500
ascent on the raw score, step $1/(n+1)$	33,275	0	492.9

“Absorbed” means the step landed exactly on 0 or 1, where the score is undefined and no further observation can move the value. `fibres.py`, experiment E2.

The middle row deserves a comment, because it is the bootstrap trap of §8 arriving by a different road. The explicit step with unit size is $s + (x - s)/n$, which at $n = 1$ is exactly x : the learner's first observation pins it to an endpoint, and an endpoint is absorbing. The implicit step never reaches an endpoint because it always retains weight $n/(n + 1) > 0$ on where it was. Stability and the avoidance of dead hypotheses are the same property.

6.3 What has been derived

Principle 6.7 (Confidence is the metric, not a second value). The resolution semantics of §3 needs a single number per candidate, and it gets one: the clause is valued at the power sc . The learning dynamics needs, in addition, the geometry in which that number is to be moved, and that geometry is a scalar field on the fibre — the Fisher information, which is the evidence count in disguise. So a PLN truth value is not a two-dimensional value. It is a one-dimensional value together with the metric at that point. Grading resolves; the metric learns.

This settles, in the direction of the second option, a question the earlier note left as a concession. There the observation was that PLN's revision is not a semiring addition — it is not idempotent, has no unit and does not distribute — and the recommendation was that $\mathbb{R}_{\geq 0}$ supply the algebra while PLN supplied values. That looked like a compromise. It is not: revision was never a candidate for the algebra, because it is not an operation on values at all. It is a step in a search over values, and steps in a search do not have to be semiring operations. The apparent defect was a category error, and correcting it removes the last piece of arbitrariness from the design.

Remark 6.8 (Two coordinates, one of which is not observable at resolution). The earlier note also observed that confidence can be invisible: if the branch competing with sc is the complement of the same belief, weighted $(1 - s)c$, then normalisation gives $sc/(sc + (1 - s)c) = s$ and c cancels exactly. Principle 6.7 explains this rather than merely recording it. The metric is not a value, so it does not appear in a ratio of values. It appears in the *step*, and a learner in that configuration still updates differently according to c even though it acts identically. Confidence becomes visible in behaviour only when there is an outside option — something to do other than choose between two readings of one hypothesis — because only then is the absolute magnitude of sc , rather than its ratio to itself, at stake.

7 The personality parameter is a step-size schedule

Definition 7.1 (The schedule). Along a sequence of observations the implicit step of Corollary 6.5 has size $\eta_n = 1/(n + 1)$, and if the learner begins with k units of prior evidence the schedule seen from the start is $\eta_j = 1/(j + k)$.

Proposition 7.2 (Robbins–Monro, for every k). *For every $k > 0$, $\sum_{j \geq 1} \eta_j = \infty$ and $\sum_{j \geq 1} \eta_j^2 < \infty$.*

Proof. $\sum 1/(j + k)$ diverges by comparison with the harmonic series and $\sum 1/(j + k)^2$ converges by comparison with $\sum 1/j^2$; neither comparison depends on k . $\square$

Empirical finding 7.3 (The schedule, measured). Truncating at $N = 200,000$:

k	$\sum_{j < N} \eta_j$	$\sum_{j < N} \eta_j^2$
1	11.7833	0.644929
4	10.7000	0.221318
9	9.9544	0.105161
64	8.0397	0.015499
512	5.9693	0.001946

The first column grows like $\log N$ for every k ; the second is bounded. `fibre.py`, experiment E3.

Corollary 7.4 (k is a transient parameter). *Under the standard stochastic-approximation hypotheses the schedule converges for every k , so k does not determine whether the learner arrives, only how it travels. An immortal learner is therefore indifferent to k .*

Principle 7.5 (Only mortality can price the learning rate). Since every k converges, no purely statistical criterion selects one. A learner with a finite stack that can be exhausted during the transient is a criterion: it is harmed by excursions it does not survive, and rewarded for opportunities it takes early enough to profit from. So the choice of k — which in the probabilistic-logic literature is set by taste, and named for taste — acquires a selection criterion exactly when the learner is embedded in an ecology that can kill it.

This is the point at which the traffic runs both ways. The framework gains a principled derivation of PLN; PLN gains a criterion for the one parameter it could not otherwise justify. §9 measures it.

8 The bootstrap trap is a vanishing gradient

Proposition 8.1 (Zero is absorbing). *Suppose a learner’s only way of acquiring evidence about ψ is to fire the receipt guarded by ψ , and suppose ψ evaluates to 0 on every candidate. Then by Definition 2.6 the receipt offers no transition, so it never fires, so no evidence arrives, so ψ is never revised. The state is a fixed point of the learning dynamics, and it is not a local optimum of anything.*

Corollary 8.2 (The trap, concretely). *With $c = n/(n + k)$ and $n_0 = 0$ we get $c = 0$, hence $\text{pow} = \text{sc} = 0$, hence a continuously enabled receipt that never fires. A learner initialised with no prior evidence at all does not die; it does nothing.*

Empirical finding 8.3 (Stasis, measured). In the baseline ecology of §9, the arm initialised at $n_0 = 0$ opens nothing, ends with s unchanged at 0.5000, and terminates with a mean stack of 520.0, which is exactly the value of grazing and never acting. The same learner with one unit of prior evidence reaches a mean terminal stack of 716.7. Survival in the first case is 1.0000: the trap is not lethal, which is what makes it hard to see.

In gradient vocabulary this is a dead unit, and the diagnosis is the standard one: the score is undefined where the branch probability is zero, so the learner sits at a point at which its own gradient does not exist. What is unusual is that the framework supplies both standard remedies *inside the clause language*, so that neither is a training-loop artefact bolted onto the semantics.

Definition 8.4 (Two repairs, both written as formulae). *Initialisation*: require $n_0 \geq 1$, which is prior evidence and is stated as part of the hypothesis. *Exploration*: replace ψ by $\psi \oplus \varepsilon$ for a constant $\varepsilon > 0$, a graded disjunction guaranteeing a floor on the weight.

Principle 8.5 (Exploration is a disjunct). The exploration bonus of a reinforcement learner is, in this setting, the statement that the learner will act if its hypothesis says so *or* for no reason. It is a connective of the hypothesis language, not a parameter of a training procedure, and it is priced like any other disjunct: the ε branch fires at a rate the ecology charges for.

Remark 8.6 (The other end of the same axis). The trap is what happens as the weight tends to zero. §9 exhibits the opposite end, where a very large k suppresses the weight almost as effectively and the learner’s behaviour becomes nearly indistinguishable from the trapped one. Between them lies whatever interior optimum the ecology allows, and part of the value of measuring it is finding out how narrow the interval is.

9 The price of the transient, measured across an ecology

9.1 Why the sweep is over distributions and not over payoffs

An earlier version of this material fixed a payoff table — a probability that a specimen is rich, a pair of likelihoods for the cheap test, two stack sizes — and varied one number at a time. A referee’s objection to that was correct and worth restating: a synthetic decision problem in which a tunable parameter develops an interior optimum shows that *this* decision problem has one. It establishes nothing general.

The repair is not to add more payoff tables but to change what is being varied. The companion programme does not describe environments by payoffs. It describes them by three distributions, and those three are what specify an ecology:

(T) the distribution over trophic types. Among the computations that can be opened at all, some are *autotrophs* — they draw only on free reservoirs and have accumulated little; some are *heterotrophs* — they eat other computations, and have therefore accumulated much; and some are composites, which do both. The trophic type is an access profile, not a species: it is a fact about what a computation’s cut permits.

(I) the distribution over inedibles. Computations that cannot be opened at a profit. They are not decoys planted by an experimenter; they are the ordinary residue of an ecology in which most things are not food.

(R) the distribution over token reservoirs housed in no trophic type. Free stacks — the wall. Non-exclusive, rate-limited, and the learner’s outside option.

Varying these is varying the ecology rather than the puzzle, and the question “does the interior optimum survive?” becomes a question about which ecologies support learners of which temperament.

9.2 The setting

A forager senses at a price $\kappa_{\text{sense}} = 4$, meets a free reservoir with probability R yielding 5, and otherwise meets a computation drawn from the composition (T, I) . It observes the cheap structural predicate χ , the presence of harvesting apparatus, whose likelihood varies by type:

$$\Pr(\chi \mid \text{hetero}) = 0.85, \quad \Pr(\chi \mid \text{comp}) = 0.75, \quad \Pr(\chi \mid \text{auto}) = 0.45, \quad \Pr(\chi \mid \text{inedible}) = 0.30.$$

Holdings are $\sigma = 130, 70, 30, 0$ for the four kinds, and breaking in costs $\kappa_{\text{break}} = 60$; so heterotrophs and composites are profitable and autotrophs and inedibles are not. The learner’s hypothesis is $\text{Inheritance}(\chi, \text{Profitable})$, held as (s, n) and placed in the **where** clause of the break-in receipt at the power sc , racing the graze branch at fixed weight $g = 0.5$. Endowment 120, horizon 400. Two openings of headroom: an early mistake is close to fatal, which is what puts the transient of §7 on the critical path.

Baseline ecology: $T = 0.40$, $I = 0.40$, $R = 0.25$,
 $\Pr(\text{Profitable} \mid \chi) = 0.6278$, $E[\text{net} \mid \text{open}] = +11.11$

arm	survival	wealth	log-fitness	final s
naive ($n_0 = 0$)	1.0000	520.0	6.2558	0.5000
crisp	0.4384	626.2	3.1374	0.4801
individual (n capped)	0.6851	604.5	4.5755	0.5661
graded, $k = 4$	0.6128	716.7	4.2652	0.5313

The crisp arm is the note as it stood before grading: a Boolean guard that fires whenever χ holds. It has the highest wealth conditional on surviving and much the worst survival, which is the earlier finding reproduced in a different world — a Boolean guard has no coordinate in which to record disappointment, so it cannot learn to stop. The naive arm is the bootstrap trap.

9.3 Two objectives, side by side

k	survival	wealth	log-fitness
0.5	0.5122	646.0	3.6037
1	0.5283	658.4	3.7116
2	0.5512	676.0	3.8625
4	0.6082	711.0	4.2345
9	0.7133	744.6	4.8794
16	0.8260	738.5	5.5164
32	0.9537	670.6	6.1568
64	0.9952	575.0	6.2869
128	0.9998	536.4	6.2717
256	1.0000	525.3	6.2612
512	1.0000	522.2	6.2580

Empirical finding 9.1 (The two criteria select different learners). In the baseline ecology the optimum under expected terminal stack is $k^* = 9$, with survival 0.7133; the optimum under expected log terminal stack is $k^* = 64$, with survival 0.9952. The two differ by a factor of seven, and the learner preferred by one criterion is killed three times as often as the learner preferred by the other. `ecology.py`.

Neither criterion is obviously wrong, and it is worth being explicit about what each is for. Expected terminal stack is the right question if the learner is one draw from a population whose aggregate holdings matter and whose individual deaths do not. Expected log terminal stack is the right question if the learner is a single lineage in a multiplicative process with an absorbing barrier, which is what a mortal computation with a conserved stack actually is. The framework does not adjudicate; it makes the disagreement visible and prices both sides.

Empirical finding 9.2 (The criteria disagree about whether learning pays at all). Under expected wealth the best graded forager reaches 744.6 against the do-nothing forager’s 520.0, a gain of 43%. Under log-fitness the best graded forager reaches 6.2869 against the do-nothing forager’s 6.2558, a margin of 0.031 — and every $k \geq 128$ scores below 6.2717, converging on the do-nothing value from above. In this ecology the log criterion can barely distinguish a learner from a learner that never acts.

That finding is the honest centre of this section, and it sharpens Remark 8.6. The trap at $n_0 = 0$ and the trap at $k \rightarrow \infty$ are the same trap approached from two directions, and under a

criterion that rewards survival above all, both look excellent. A framework that reports only log-fitness would conclude that the best policy is to do nothing, and it would be right about survival and wrong about everything else.

9.4 The three sweeps

Each sweep varies one distribution and holds the others at the baseline. “ $k^*(w)$ ” is the optimum under expected wealth, “ $k^*(\ell)$ ” under log-fitness, each with the survival it achieves; 3000 lives per cell, eleven values of k per cell.

(T) the distribution over trophic types: heterotroph share among edibles

T	$\Pr(\text{Prof} \mid \chi)$	$k^*(w)$	survival	$k^*(\ell)$	survival	$E[\text{net}]$
0.00	0.4687	512	1.0000	512	1.0000	−18.75
0.20	0.5529	512	1.0000	512	1.0000	−2.94
0.40	0.6278	9	0.7123	64	0.9943	+11.11
0.60	0.6947	4	0.8223	16	0.9313	+23.68
0.80	0.7550	1	0.8507	16	0.9620	+35.00
1.00	0.8095	2	0.9117	9	0.9633	+45.24

(I) the distribution over inedibles

I	$\Pr(\text{Prof} \mid \chi)$	$k^*(w)$	survival	$k^*(\ell)$	survival	$E[\text{net}]$
0.00	0.8071	0.5	0.9683	4	0.9860	+31.43
0.20	0.7290	1	0.8297	16	0.9610	+22.58
0.40	0.6278	9	0.7123	64	0.9943	+11.11
0.60	0.4913	512	1.0000	512	1.0000	−4.35
0.80	0.2974	512	1.0000	512	1.0000	−26.32

(R) the distribution over free reservoirs

R	$\Pr(\text{Prof} \mid \chi)$	$k^*(w)$	survival	$k^*(\ell)$	survival	$E[\text{net}]$
0.00	0.6278	16	0.7267	128	0.9983	+11.11
0.15	0.6278	9	0.6807	128	0.9993	+11.11
0.30	0.6278	9	0.7463	64	0.9977	+11.11
0.45	0.6278	9	0.8300	32	0.9870	+11.11
0.60	0.6278	9	0.9147	32	0.9940	+11.11

9.5 Interaction between two of the distributions

The sweeps above vary one distribution at a time. Because (T) and (I) both enter the conditional $\Pr(\text{Prof} \mid \chi)$, they might be expected to interact, and a coarser grid over the pair confirms that they do — and that the interaction is legible. Cells use 1200 lives, so the optima are noisier than in the sweeps; the baseline cell reports $k^*(w) = 16$ here against 9 at the higher count.

T	I	$\Pr(\text{Prof} \mid \chi)$	$E[\text{net}]$	$k^*(\ell)$	$k^*(w)$
0.15	0.15	0.646	+4.6	512	512
0.15	0.40	0.533	−6.7	512	512
0.15	0.70	0.334	−26.6	512	512
0.40	0.15	0.750	+25.0	16	1
0.40	0.40	0.628	+11.1	64	16
0.40	0.70	0.404	−14.3	512	512
0.75	0.15	0.870	+48.4	9	1
0.75	0.40	0.741	+32.3	16	2
0.75	0.70	0.492	+1.3	512	512

9.6 What the surface says

Empirical finding 9.3 (The interior optimum requires a margin, not merely a profit). Across all 25 cells reported above, the two optima agree about which side of the grid they fall on. Every cell with $E[\text{net} \mid \text{open}] \geq +11.1$ has an interior optimum under both objectives; every cell with $E[\text{net} \mid \text{open}] \leq +4.6$ has a boundary optimum under both. Sixteen cells are interior and nine are at the boundary, and no cell in the measured range lies between +4.6 and +11.1.

Two of the boundary cells — $(T, I) = (0.15, 0.15)$ at +4.6 and $(0.75, 0.70)$ at +1.3 — have *positive* expected net yield from opening and still send the personality parameter to maximal scepticism. That is the finding worth having, and it is stronger than the one anticipated. Positive expectation is not sufficient to make acting worthwhile for a learner that can die. With two openings of headroom, a thin positive mean is dominated by the variance around it, and the correct policy is abstention. The threshold is therefore strictly positive, and where it sits is a joint property of the ecology’s payoffs and the learner’s endowment — which is the Kelly criterion arriving from the ecological side rather than the financial one.

This is also the correct and modest version of the earlier claim about an interior optimum. There is no general interior optimum for the personality parameter. There is one exactly when the ecology makes acting worth doing by a margin, and where it does not, maximal scepticism is not a failure of nerve but the right answer — the boundary optimum is the framework saying so. What survives robustness is not the number but the criterion.

Empirical finding 9.4 (Scepticism and free energy are substitutes). Along the (R) sweep the hypothesis being learned does not change: the truth $\Pr(\text{Prof} \mid \chi)$ is 0.6278 in every cell, because free reservoirs are not computations and do not enter the conditional. Yet $k^*(\ell)$ falls from 128 to 32 as reservoirs become abundant, and survival at the wealth optimum rises from 0.6807 to 0.9147.

Principle 9.5 (The transient is what an outside option pays for). A rich outside option does not make the learner’s hypothesis truer or its test sharper. It makes the transient survivable, by supplying income the learner does not have to gamble for. Scepticism is the other way of surviving a transient, so the two trade off, and an ecology with abundant free energy can afford credulous learners. This is a statement about environments, not about learners, and it is statable only because the framework prices perception, action and survival in one currency.

Remark 9.6 (What is not claimed). These are measurements on a constructed ecology with four kinds of computation, a single binary structural predicate, and a fixed price schedule. What has been established is the shape of the dependence — which way k^* moves as each distribution moves, and where the interior optimum stops existing — not a quantitative law. The numbers are reproducible from `ecology.py` and should be read as the behaviour of that model.

10 Where the gradient cannot go

Everything in §§5–9 happens in one fibre. The learner tunes θ , and θ is the value of a hypothesis whose *shape* it never questions. It is worth being explicit about how severe that restriction is, because the machinery is seductive and the restriction is not visible from inside it.

Proposition 10.1 (The fibre gradient never changes the theory). *Let $(\psi, \theta) \mapsto (\psi, \theta')$ be any sequence of fibre moves. Then the base component is constant, so by Proposition 4.1 the ultrametric distance travelled is 0, and by Proposition 4.2 the learner remains in every ball it started in — including the ball of radius 0.*

A learner equipped only with gradient descent is confined absolutely. It is not slow to escape a bad theory; it cannot form the question. The forager of §9 is the illustration and it is a sharp one: it learns beautifully not to open inedibles, and in four hundred steps it never once considers that a second structural predicate might separate heterotrophs from autotrophs, which is the hypothesis that would actually pay.

Principle 10.2 (What the fibre buys, exactly). Before grading, the only way to change behaviour was to change the formula: a discrete jump, priced by backtracking depth, invalidating every confirmation obtained below that depth. After grading, behaviour can be changed continuously while the formula and the evidence are held fixed. Grading does not enlarge the set of reachable theories by one element. It makes the interior of a ball navigable, and §9 measures what that is worth: in the harsh cells it is the difference between abstention and extinction.

10.1 What escapes

Three operators move in the base, and all three are structural rather than epistemic. They are treated in the companion notes and are recalled here only to place them relative to the gradient.

Recombination. Reproduction by crossover on quoted code is a base move of arbitrary size. Because a germ is a quotation — inert, transmissible, inspectable — crossover can exchange whole subformulae, which is precisely the move an incremental revision walk cannot make. The transport claim of the scientist note, that a revision move and a crossover are one move carried along the characteristic-formula map and distinguished only by accessible distance, is what makes this a statement about the same space rather than an analogy.

Composition. Two learners in parallel constitute a learner whose namespace is the union, and disjointness of namespaces is exactly probabilistic independence of their distributions. Composition therefore reaches shapes neither component could reach, at the price of the interaction term.

Death. A population in which lineages end is a search that can abandon a ball, which no individual can do.

Principle 10.3 (The division of labour is forced by the geometry). Continuous parameter learning and discrete structure search are not two design choices made by an engineer who could have made others. They are what the two components of a fibration respectively admit: an ordered continuum has nothing but gradients, and an ultrametric has none. A framework in which hypotheses are formulae and confidences are numbers must have gradient descent on the numbers and evolutionary or backtracking search on the formulae, and it must have both because neither reaches what the other reaches.

That this reproduces the familiar arrangement — stochastic gradient descent over parameters, discrete search over architectures — is worth saying plainly, because the point is the direction of the derivation. The arrangement is not imported here as good practice. It is a consequence of the geometry of a hypothesis space whose points are formulae, and the framework predicts it rather than assuming it.

11 Testimony, and the conservation of evidence

11.1 Two conservation laws, one of which is false

Tokens are conserved. The game rules of the companion programme are explicit about it: a stack may be moved, split, or spent, and never minted, and the whole economics of the mortal scientist rests on that.

Evidence is not conserved. Nothing in the framework prevents a learner from counting the same observation twice, and PLN revision — which by Theorem 6.3 is the fibre’s gradient step — presumes that the evidence it is handed is independent of the evidence it already has. Two learners who both heard from a third and then revise by each other’s truth values have double-counted, and the arithmetic cannot tell.

Principle 11.1 (Belief inflation is minting). The metabolic namespace has a conservation law and the belief namespace does not. So the epistemic analogue of counterfeiting exists, has no metabolic analogue, and requires no defect in any participant: honest agents running the correct update rule on correlated evidence inflate the count.

11.2 What inflation does, in gradient terms

The natural reading of counterfeit confidence is recklessness — an agent that believes too strongly and acts too boldly. Theorem 6.3 says the opposite, and the prediction is sharp enough to test.

Proposition 11.2 (Inflation collapses the step size). *The evidence count is the Fisher information up to the variance factor, and the step size is $\eta = 1/(n + 1)$. Inflating n by a factor λ therefore divides the learner’s step size by approximately λ . A learner with counterfeit confidence is not unstable. It is rigid.*

Corollary 11.3 (The damage is invisible in a stationary world). *If the truth being estimated does not move, a collapsed step size costs almost nothing: the learner has already converged to approximately the right value and is merely unable to move away from it. The cost of minted confidence appears only when the environment changes, and it appears as an inability to re-adapt.*

11.3 The experiment

Two learners share a belief namespace. Each forages a stretch of the horizon and then, in some protocols, exchanges truth values with the other. Three protocols:

isolated no exchange;

honest one exchange at the midpoint, of evidence drawn from disjoint streams, so counts add and the sum is warranted;

plagiarism exchange at every epoch, so that each learner’s count is re-transmitted and re-added on the following exchange, and the same observations circulate.

The ecology is the baseline of §9 with $k = 9$, run in two regimes: one in which the trophic composition shifts at the midpoint from $(T, I) = (0.40, 0.40)$ to $(0.05, 0.65)$, moving the truth from 0.6278 to 0.3409, and a stationary control. “Inflation” is the ratio of the nominal evidence count to the count actually warranted by distinct observations.

Shifting ecology: truth 0.6278 $\rightarrow$ 0.3409 at the midpoint

protocol	inflation n/n_{indep}	$ s - \text{truth} $ pre-shift	$ s - \text{truth} $ post-shift	step $1/(n + 1)$	survival	wealth
isolated	1.000	0.1246	0.1573	0.05295	0.1590	68.4
honest	1.000	0.0706	0.2052	0.02183	0.1307	62.2
plagiarism	140.98	0.0752	0.2446	0.00043	0.1183	61.7

Stationary control: truth 0.6278 throughout

protocol	inflation n/n_{indep}	$ s - \text{truth} $ first half	$ s - \text{truth} $ second half	step $1/(n + 1)$	survival	wealth
isolated	1.000	0.1197	0.0982	0.04213	0.7350	772.3
honest	1.000	0.0703	0.0583	0.01815	0.7213	766.2
plagiarism	133.52	0.0747	0.0699	0.00042	0.6637	754.6

11.4 Four findings

Empirical finding 11.4 (Honest testimony is superadditive, and cheap). One exchange of independent evidence reduces the error in s from 0.1246 to 0.0706 in the shifting run and from 0.1197 to 0.0703 in the control — close to the factor $\sqrt{2}$ that doubling independent evidence predicts — at no metabolic cost to either party beyond the message. This is the inequality behind the slogan that cooperation is epistemic and competition metabolic, with a number attached.

Empirical finding 11.5 (Plagiarism buys accuracy nothing and count everything). The plagiarism protocol reaches pre-shift error 0.0752, no better than the single honest exchange’s 0.0706, while its nominal evidence count is inflated by a factor of 140.98. Confidence has been minted at a ratio of one hundred and forty to one against the evidence backing it.

Empirical finding 11.6 (The pathology is rigidity, and it is invisible until the world moves). Step sizes at the end of the run are 0.05295, 0.02183 and 0.00043 for the three protocols. In the stationary control the ordering of post-period accuracy follows accuracy, as one would hope: honest 0.0583, plagiarism 0.0699, isolated 0.0982. In the shifting run the ordering *inverts* and follows step size instead: isolated 0.1573, honest 0.2052, plagiarism 0.2446. The plagiarists are the least able to notice that the world has changed, exactly as Proposition 11.2 predicts, and the honest sharers are second-least able.

The second half of that finding is the uncomfortable one and should not be softened. Under a regime shift, *any* accumulation of evidence is a liability, and honest testimony is a liability too — smaller than counterfeit testimony, larger than isolation. The Robbins–Monro schedule is designed for a stationary target. A mortal learner in a changing ecology is not solving that problem, and the framework makes the mismatch measurable rather than concealing it. The obvious repair, discounting old evidence so that η has a floor, is not innocent either: it is a step size that does not satisfy Proposition 7.2, so it trades convergence for plasticity.

Empirical finding 11.7 (Accuracy and fitness come apart, with a sign). In both regimes the metabolic ordering is the reverse of the epistemic one: isolated learners hold the most wealth (772.3 in the control) and the most accurate learners the least (754.6 for plagiarism, 766.2 for honest exchange), and survival follows wealth. In a marginal ecology — one where $E[\text{net} \mid \text{open}]$ is positive but thin — converging quickly to the true strength means acting on it, and acting on a thin edge is expensive. The earlier programme’s slogan that a learner is paid in yield and not in truth is reproduced here independently, and with the additional detail that testimony sharpens truth while slightly degrading yield.

Remark 11.8 (What the two-learner run does not settle). The protocols here are stipulated rather than chosen by the learners. Whether a population in which testimony is available has an equilibrium, and whether plagiarism invades it, requires the belief to be a strategy and the exchange to be priced — which is composition in the sense of the companion note, and is left open. Nor is any defence attempted: an evidence-provenance discipline that made double counting detectable would be the belief-namespace analogue of the conservation law the metabolic namespace already has, and the framework says clearly what it would have to do, but not how.

12 Implementation

Three pieces of what is described here already exist, one does not, and the boundary is worth being precise about.

The clause. The **where** clause is present in the reference node as field **condition** of the receive form, evaluated in the matcher by a separate pure-evaluation crate. Per-match evaluation is what the matcher already does, since a guard must be checked against the bindings a candidate supplies. Making the clause $\mathbb{R}_{\geq 0}$ -valued rather than Boolean is a change of codomain in one function, and the classical interpreter can continue to read only the support — fires if non-zero — so a graded program remains a legal classical program.

The simulator. The weighted-GSLT crate carries rules equipped with weight maps keyed by spatial-behavioural formulae, computes propensities by redex, and runs both a classical stochastic simulation and a quantum one. Its weight maps are, by the normal-form theorem of Paper III, graded clauses in disjunctive form, so it is already an implementation of the special case. What it does not have is a clause language: the keys are a hand-rolled formula enum and the partition discipline it enforces is exactly the discipline a single-valued clause makes unnecessary.

The gradient. Corollary 5.8 says what a runtime would have to carry, and it is little. At each firing, the site already computed the weight of every candidate in its contention set in order to resolve the race. To accumulate the score it must additionally compute, for each parameter, the derivative of each candidate weight — available in closed form from Proposition 5.2, since a posynomial’s derivative is a posynomial — and add one number per parameter to a running total. That total is an eligibility trace, and in **rho** the natural place for it is a message on a channel private to the learner, which makes it harvestable like any other belief. Nothing requires a backward pass over the reduction graph, and nothing requires the graph to be retained.

The belief. Because a truth value sits on a channel as data, revision is an ordinary communication and Remark 2.7 does the rest. The consequence noticed in the earlier note stands and is sharpened by §11: a belief is a message, so testimony and the theft of a hypothesis are the same operation as predation, differing only in which namespace the name lives in.

What is not built. None of the gradient machinery. The scripts shipped with this note are simulations of the mathematics, not instrumentation of a node: **fibre.py** verifies the identities and **ecology.py** runs the foragers, and neither executes **rho**.

13 Related work

Semiring-weighted computation. Weighting the transitions of a system by elements of a semiring, multiplying along paths and summing across them, is long established — weighted automata are the canonical instance, and semiring-valued logics predate this line of work by decades. Nothing in the algebra of §3 is new, and the earlier version of this material was rightly criticised for not saying so.

Stochastic process calculi. Rates attached to actions, with races resolved by relative rate and holding times drawn from the total, are the design of the stochastic pi calculus and its machine, and of the performance-evaluation calculi. The differences here are two. The weight is attached to a *candidate match* rather than to a channel or an action, so it may depend on the datum; and it is written as a formula of a logic generated by the term language rather than as a numeral, which is what makes it something a learner can hold as a hypothesis.

The novelty claim. Stated narrowly, and following a reviewer’s own sharpening: placing a semiring-valued expression exactly at rho-calculus candidate-match resolution. What this paper adds to that claim is the reading of the resulting parameter as the fibre of a fibration over the hypothesis lattice, and the identification of the metric on that fibre.

Probabilistic logic networks. PLN supplies the truth-value structure of §6: a strength, an evidence count, a confidence $n/(n + k)$ with a personality parameter, and revision by count-weighted averaging. This paper’s contribution to PLN, if it is one, is Theorem 6.3 and Principle 7.5: the second coordinate is derivable as the Fisher metric rather than postulated, and k acquires a selection criterion it does not have on statistical grounds alone.

Conjugate updating. A pair (s, n) is a summary of a Beta–Bernoulli posterior and revision is conjugate updating, which is older than either PLN or this note and should be said plainly. Two things are not supplied by conjugacy. One is the calculus: PLN’s contribution is a set of composition rules for truth values under inference, and it is those, not the update, that make a truth value something a formula can carry. The other is the observation that the conjugate update coincides with a natural-gradient step, which is what transports the construction from inference to *search* — from what a learner believes to how it moves.

Stochastic approximation and natural gradient. The step-size conditions of Proposition 7.2 are the classical Robbins–Monro conditions; the metric of Definition 6.1 is Amari’s natural gradient; the estimator of Theorem 5.6 is the likelihood-ratio or score-function estimator, and combined with a terminal payoff it is policy-gradient reinforcement learning. Implicit and proximal discretisations of gradient flow, and their unconditional stability, are standard in numerical analysis. What is claimed here is not any of these results but their location: each arrives as a consequence of the semiring and the operational semantics rather than as a modelling choice.

Kelly. The log criterion of §9 is the growth-optimal criterion, and Finding 9.3’s margin requirement is its ecological shadow. The framework’s contribution is that the two criteria can be run against each other in one currency, since perception, action and survival are all priced in tokens.

14 What is shown, what is not, and what is open

14.1 Shown

1. Over $\mathbb{R}_{\geq 0}$ with positive constants, support adequacy is an equality, so every reachable branch has a probability (Proposition 3.4).
2. A modality-free clause is a posynomial in its parameters, hence smooth and log-convex in log-coordinates; its normalisation is not (Propositions 5.2, 5.3; measured, Finding 5.4).
3. The parameter gradient of a causal history’s log-likelihood is a sum of local scores (Theorem 5.6; measured, Finding 5.7).
4. The gradient of a modal clause is the solution of the adjoint system, whose Neumann series is that of the fixed point itself, so differentiability and existence hold under one condition (Theorem 5.12, Corollary 5.13; measured, Finding 5.14).
5. PLN revision is the natural-gradient step with unit step size and the Fisher metric at the posterior count (Theorem 6.3; measured with a negative control, Finding 6.4).
6. That step is implicit, hence never leaves the unit interval, while the explicit forms are absorbed at the boundary or escape it (Corollary 6.5; measured, Finding 6.6).
7. The confidence schedule is Robbins–Monro for every personality parameter, so k governs the transient (Proposition 7.2, Corollary 7.4).
8. A clause valued at zero is a fixed point of the learning dynamics that is not an optimum, and both standard remedies are expressible as formulae (Proposition 8.1, Definition 8.4).
9. Fibre moves have base-distance zero, so a purely graded learner is confined absolutely (Proposition 10.1).

14.2 Measured but not proved

The robustness surface of §9 — in particular Finding 9.3, that an interior optimum for k requires a strictly positive margin and not merely a positive expectation — is an empirical statement about one constructed ecology with four kinds of computation and one binary predicate. The threshold’s dependence on the endowment is conjectural: the Kelly reading predicts that it scales with the ratio of κ_{break} to the endowment, and that is not tested here.

Likewise Finding 11.6: the inversion of the accuracy ordering under a regime shift is measured on one shift, of one magnitude, at one time.

14.3 Not shown

The gradient is never run in a `rho` node; the scores of Theorem 5.6 are computed by a simulator that enumerates candidates, not by a runtime that fires them. No claim is made that a learner using this machinery outperforms one that does not on any task outside the constructed ecology. And the two-sorted grammar’s crisp side is taken as given: what makes a generated logic adequate, and what its target specification must supply for negation to exist, is Paper III’s business and the omnibus’s.

14.4 Open

Obligation 14.1 (Non-stationary schedules). Proposition 7.2 and Finding 11.6 are in tension: the schedule that guarantees convergence is the one that destroys plasticity. A discounted count, giving η a floor, breaks Robbins–Monro. Whether a mortal learner should want convergence at all — and what the right criterion is when the target moves at a rate the learner could in principle estimate — is open, and it is the most consequential gap in this paper.

Obligation 14.2 (Provenance, and a conservation law for evidence). Section 11 exhibits minting and does not prevent it. A provenance discipline making double counting detectable would be the belief-side analogue of token conservation. The natural candidate — evidence tagged by the quotation of the observation that produced it, so that revision may deduplicate — is cheap to state and expensive to run, since it makes a truth value grow with its history.

Obligation 14.3 (The base, priced against the fibre). Principle 4.5 says a learner needs both searches and that they are priced differently. It does not say how a learner should divide a finite stack between them. The two prices are in the same currency, so the question is well posed and unanswered.

Obligation 14.4 (Second-order structure). The Fisher metric is available on the fibre by Theorem 6.3 for a single Bernoulli parameter. For a clause with d parameters the metric is a $d \times d$ matrix, and whether it retains the closed form — whether, that is, the evidence counts of several parameters compose as simply as one — is not settled here. Proposition 5.2 suggests it should, since a posynomial’s log-Hessian is explicit.

Obligation 14.5 (The holding times). Remark 3.3 confines the graded semantics to the jump chain. A learner tuning clause values cannot therefore tune *when* it acts, only what it does when it acts. Whether the waiting-time law should also be graded, and what a learner would then be learning, is a question for the weighted-GSLT companion.

15 Conclusion

The rho calculus is silent about how a race is resolved, and this line of work treats the silence as a parameter. Paper I sets the parameter to the complex numbers and asks what has to be true of a program before its denotation is a quantum state. This paper sets it to the non-negative reals and asks a different kind of question, because over the reals the semantics is unremarkable and the learner is the point.

The answer has a shape that was not obvious at the outset. Grading does not give a mortal scientist a better hypothesis space; the hypothesis space is an ultrametric tree and remains one, and confinement is still a theorem. What grading gives it is a second coordinate in which small moves mean something, and the resulting structure is a fibration: backtracking in the base, gradient in the fibre, and no move of either kind substituting for the other. That the familiar division of machine learning into architecture search and parameter fitting falls out of this rather than being assumed is the most portable thing here.

The technical centre is smaller and sharper. Asking for the gradient in the fibre to be taken correctly forces a metric; the metric is the Fisher information; the Fisher information for a strength is an evidence count; and a strength together with an evidence count is a probabilistic-logic truth value, whose revision rule is then not a rule at all but the implicit discretisation of the flow. So confidence is not a second thing a learner believes. It is the shape of the space in which its belief is allowed to move, and the reason its steps get smaller is that they are supposed to.

What the ecology adds is a price. Every step size converges, so statistics cannot choose one; a learner that can die during the transient can, and does. The measurements say that the choice is real, that the two obvious objectives make it differently and disagree by a factor of seven, that an interior answer exists only where the ecology pays a margin rather than merely a profit, and that a rich outside option substitutes for caution. They also say something less comfortable: that a criterion which rewards survival above all cannot easily tell a learner from a learner that never acts, and that under a moving target the accumulation of evidence — honest evidence, correctly handled — is itself a liability. The framework does not resolve these. It makes them arithmetic, which is the most a framework can do.

A The forager, in rholang

The listing is the assay of §9 written in the surface syntax, with the belief resting on a channel so that revision is an ordinary communication. The clause is the only graded expression; everything else is rho 1.4.

```
new belief, spec, wall, sense, revise in {
  // the belief is a message: a strength and an evidence count.
  // k is the personality parameter of Definition 7.1.
  belief!(0.5, 1) |

  // the assay. Two receipts contend for one specimen, so they form a
  // single contention set and their weights are normalised against each
  // other. The break-in branch is valued at the power s*c; the graze
  // branch carries the constant g.
  contract sense(@specimen, @k, @g) = {
    for( @(s, n) <- belief ) {
      belief!(s, n) |
      for( @x <- spec & @chi <- specimen
        where s * n / (n + k) )
        { revise!(*x) }
      |
      for( @x <- spec & @chi <- specimen
        where g )
        { wall!(0) }
    }
  } |

  // revision. One observation, count one: the implicit natural-gradient
  // step of Corollary 6.6, written as arithmetic on the message.
  contract revise(@outcome) = {
    for( @(s, n) <- belief ) {
      belief!( (n * s + outcome) / (n + 1), n + 1 )
    }
  }
}
```

Three things the listing exposes. First, the entire graded content is the two **where** clauses, and they differ only in which weight they carry: the whole policy is two expressions. Second, **revise** contains no learning rate — the step size is $1/(n+1)$ and it is read off the belief rather than configured, which is Corollary 7.4 in the surface syntax. Third, the belief is a message on a channel, so any computation holding that name can read it, revise it, or take it; §11 is what happens when two of them do.

B The computations

Two scripts, standard library only, matching the convention of the companion notes.

fibre.py verifies the mathematics of §§5–7. E1: revision against the natural-gradient step, 20,000 instances, with the metric placed at the prior count as a negative control. E2: four discretisations over 200,000 steps, counting escapes from the unit interval and absorptions at its ends. E3: the Robbins–Monro sums, truncated at $N = 200,000$. E4: least eigenvalues of the log-Hessian of a clause and of a normalised branch probability, over 400 random clauses in three parameters, by cyclic Jacobi. E5: the score decomposition against central differences,

300 histories of six contention sets. E6: the adjoint derivative of a graded fixed point against central differences, 200 systems of dimension five, plus a Neumann-truncation table as the spectral radius approaches one.

`ecology.py` runs the foragers of §§9–11. It defines an ecology by the three distributions (T), (I), (R), derives $\Pr(\text{Prof} \mid \chi)$ and $E[\text{net} \mid \text{open}]$ from them rather than stipulating either, and reports both objectives at every cell. The four-arm table and the k -table use 6000–8000 lives; the three sweeps use 3000; the interaction grid uses 1200, which is why its baseline cell disagrees with the k -table’s by one grid step. The testimony experiment runs 1500 paired lives per protocol over eight epochs, tracking the nominal evidence count and the count warranted by distinct observations separately, in a shifting ecology and a stationary control.

Every number appearing in this note is produced by one of the two scripts.

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